

Day-04

Python interview questions on conditional statements:

Basic Questions:

- 1.What are conditional statements in Python?
- 2.Explain the difference between if, elif, and else.
- 3.How does Python handle multiple conditions in a single if statement?
- 4.What happens if you write an if condition without an else statement?
- 5.How do you write a one-liner if statement in Python?

Intermediate Questions:

- 1.What is the difference between if x: and if x == True:?
- 2.How do you use logical operators (and, or, not) in conditional statements?
- 3.What will be the output of this code?

code

```
x = 5
```

```
if x > 3 and x < 10:
```

```
    print("Valid")
```

- 4.How does Python evaluate expressions with multiple conditions?
- 5.What is short-circuiting in Python conditional statements?

Advanced Questions:

- 1.Can you use a conditional statement inside a list comprehension? Provide an example.
- 2.What is the difference between a ternary operator and a normal if-else statement?
- 3.Explain how match-case (introduced in Python 3.10) is different from if-elif-else.
- 4.What will be the output of the following nested if condition?

code

```
x = 7
```

```
if x > 5:
```

```
    if x < 10:
```

```
        print("Between 5 and 10")
```

```
    else:
```

```
        print("Greater than 10")
```

5.How can you avoid deep nesting in conditional statements?

Q1. Name the keyword which helps in writing code involves condition.

Ans. if

Q2. Write the syntax of simple if statement.

Ans.

```
if <Condition> :
```

```
    Execute this
```

Q3. Is there any limit of statement that can appear under an if block.

Ans. No

Q4. Write a program to check whether a person is eligible for voting or not. (accept age from user)

Ans.

```
age=int(input("Enter your age"))
```

```
if age >=18:
```

```
    print("Eligible for voting")
```

Q5. Write a program to check whether a number entered by user is even or odd.

Ans.

```
num=int(input("Enter your age"))  
if num%2==0:  
    print("Number is Even")  
else:  
    print("Number is Odd")
```

Q6. Write a program to check whether a number is divisible by 7 or not.

Ans.

```
num=int(input("Enter your age"))  
if num%7==0:  
    print("Number is divisible")  
else:  
    print("Number is not divisible")
```

Q7. Write a program to display "Hello" if a number entered by user is a multiple of five, otherwise print "Bye".

Ans.

```
num=int(input("Enter your age"))  
  
if num%5==0:  
    print("Hello")  
else:  
    print("Bye")
```

Q8. Write a program to calculate the electricity bill (accept number of unit from user) according to the following criteria:

Unit	Price
First 100 units	no charge
Next 100 units	Rs 5 per unit
After 200 units	Rs 10 per unit

(For example if input unit is 350 than total bill amount is Rs2000)

Ans.

```
amt=0
```

```
nu=int(input("Enter number of electric unit"))
```

```
if nu<=100:
```

```
    amt=0
```

```
if nu> 100 and nu<=200:
```

```
    amt=(nu-100)*5
```

```
if nu>200:
```

```
    amt=500+(nu-200)*10
```

```
print("Amount to pay:", amt)
```

Q9. Write a program to display the last digit of a number. (hint: any number % 10 will return the last digit)

Ans.

```
num=int(input("Enter any number"))
```

```
print("Last digit of number is ",num%10)
```

Q10. Write a program to check whether the last digit of a number (entered by user) is divisible by 3 or not.

Ans.

```
num=int(input("Enter any number"))
ld=num%10
if ld%3==0:
    print("Last digit of number is divisible by 3 ")
else:
    print("Last digit of number is not divisible by 3 ")
```

Q1. Write a program to accept percentage from the user and display the grade according to the following criteria:

Marks	Grade
> 90	A
> 80 and <= 90	B
>= 60 and <= 80	C
below 60	D

Ans.

```
per = int(input("Enter marks"))
if per > 90:
    print("Grade is A")
if per > 80 and per <=90: print("Grade is B")
if per >=60 and per <= 80: print("Grade is C")
if per < 60:
    print("Grade is D")
```

Q2. Write a program to accept the cost price of a bike and display the road tax to be paid according to the following criteria:

Cost price (in Rs)	Tax
>100000	15%
>50000 and <= 100000	10%
<= 50000	5%

Ans.

```
tax = 0
```

```
pr=int(input("Enter the price of bike"))
```

```
if pr > 100000:
```

```
    tax = 15/100*pr
```

```
elif pr >50000 and pr <=100000:
```

```
    tax = 10/100*pr
```

```
else:
```

```
    tax=5/100*pr
```

```
print("Tax to be paid ", tax)
```

Q3. Write a program to check whether an years is leap year or not.

Ans.

```
yr=int(input("Enter the year"))
```

```
if yr%100==0:
```

```
    if yr%400==0:
```

```
        print("Entered year is leap year")
```

```
    else:
```

```
        print("Entered year is not a leap year")
```

```
else:
```

```
    if yr%4==0:
```

```
        print("Entered year is leap year")
```

else:

print("Entered year is not a leap year")

Q4. Write a program to accept a number from 1 to 7 and display the name of the day like 1 for Sunday, 2 for Monday and so on.

Ans.

```
num=int(input("Enter any number between 1 to 7: "))
```

```
if num==1:
```

```
    print("Sunday")
```

```
elif num==2: print("Monday")
```

```
elif num==3: print("Tuesday")
```

```
elif num==4: print("Wednesday")
```

```
elif num==5: print("Thursday")
```

```
elif num==6: print("Friday")
```

```
elif num==7: print("Saturday")
```

```
else: print("Please enter number between 1 to 7")
```

Q6. What do you mean by statement?

Ans. Executable lines in a program is called instruction

Q7. Write the output of the following:

```
if True: print(101)
```

```
else: print(202)
```

Ans. 101

Q8. Write the logical expression for the following:

A is greater than B and C is greater than D

Ans. $A > B$ and $C < D$

Q9. Accept any city from the user and display monument of that city.

```
if city.lower()=="delhi": print("Monument name is: Red Fort")
elif city.lower()=="agra": print("Monument name is: Taj Mahal")
elif city.lower()=="jaipur": print("Monument name is: Jal Mahal")
else: print("Enter correct name of city")
```

Q10. Write the output of the following if $a = 9$

```
if (a > 5 and a <=10): print("Hello")
else: print("Bye")
```

11. Write a program to find the largest number out of three numbers expected from user.

```
A=input()
B=input()
C=input()
If A>B and A>C:p(A ,is greatest)
If B>A and B>C:p(B ,is greatest)
If C>B and C>A:p(C ,is greatest)
```

12. Accept the age of 4 people and display the youngest one?

13. Write a program to check a character is vowel or not .

```
Ch=input("enter any character")
Vow="aeiouAEIOU"
If ch in vow:
P("vowel")
Else:p("not vowel")
```


Q2. Accept the percentage from the user and display the grade according to the following criteria:

- Below 25 — D
- 25 to 45 — C
- 45 to 50 — B
- 50 to 60 — B+
- 60 to 80 — A
- Above 80 — A+

Ans.

```
per = int(input("Enter percentage"))
if per > 80:
    print("Grade is A+")
elif per > 60 and per <= 80:
    print("Grade is A")
elif per > 50 and per <= 60:
    print("Grade is B+")
elif per > 45 and per <= 50:
    print("Grade is B")
elif per > 25 and per <= 45:
    print("Grade is C")
elif per < 25:
    print("Grade is D")
```

Q3. A company decided to give bonus to employee according to following criteria:

Time period of Service	Bonus
More than 10 years	10%
>=6 and <=10	8%
Less than 6 years	5%

Ask user for their salary and years of service and print the net bonus amount.

Ans.

```
ser=int(input("Enter the time period of service"))
sal =int(input("Enter your salary"))
if ser > 10:
    b= 10/100*sal
if ser >=6 and ser <=10:
    b = 8/100*sal
if ser < 6:
    b = 5/100*sal
print("Bonus is ", b)
```

Q5. Write a program to accept percentage and display the Category according to the following criteria :

Percentage	Category
< 40	Failed
>=40 & <55	Fair
>=55 & <65	Good
>=65	Excellent

Ans.

```
pr = int(input("Enter the percentage"))
if pr < 40:
    print("Your Category is: Failed")
elif pr >= 40 and pr < 55:
    print("Your Category is: Fair")
elif pr >= 55 and pr < 65:
    print("Your Category is: Good")
elif pr >= 65 and pr <= 100:
    print("Your Category is: Excellent")
elif pr > 100:
    print("Please enter correct percentage")
```